ANALYSIS OF RED WINE QUALITY

Data Transformation and Predictive Modeling

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Introduction

Red Wine Quality Dataset Overview

The Red Wine Quality Dataset is derived from "RedWine.txt", used to model wine quality through physicochemical tests. It includes 1,599 samples of red wine from northern Portugal, adapted from the dataset used in the 2009 study by Cortez et al. Then, a random sample of 500 are chosen to conduct the analysis.

This dataset features six key variables:

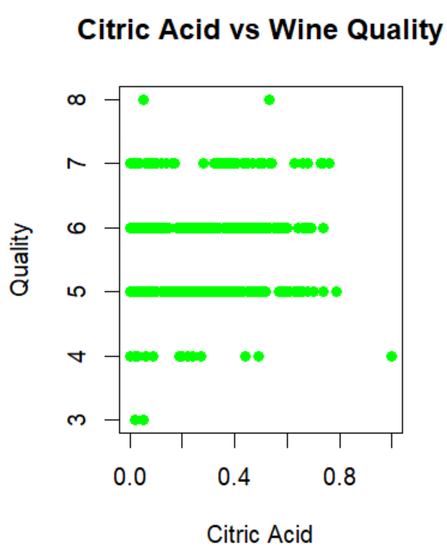
- **X1 (Citric Acid)**
- **X2 (Chlorides)**
- **X3 (Total Sulphur Dioxide)**
- **X4 (pH)**
- **X5 (Alcohol)**
- **Y (Quality)**

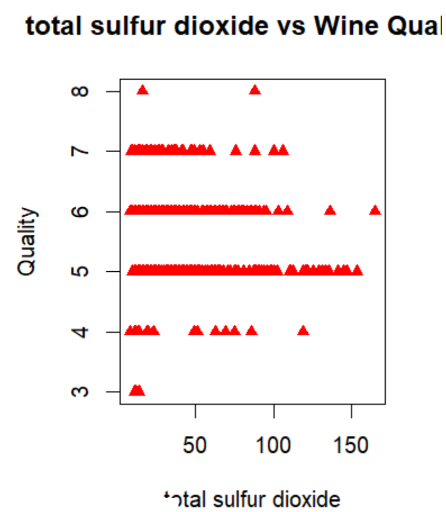
The analysis aims to explore the relationships and patterns within these variables to predict and understand the factors that most significantly influence wine quality.

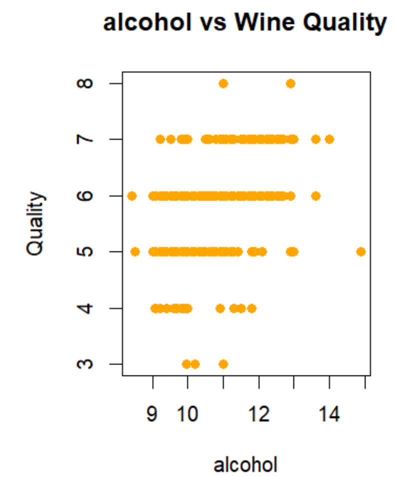
Reference:

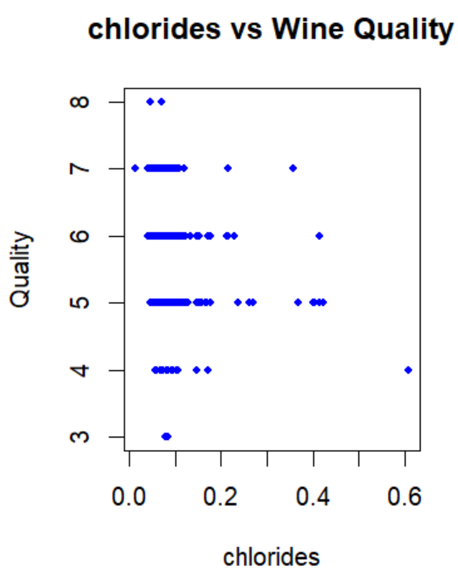
Cortez, P., Cerdeira, A., Almeida, F., Matos, T., & Reis, J. (2009). Modeling wine preferences by data mining from physicochemical properties. Decision Support Systems, Elsevier, 47(4), 547-553

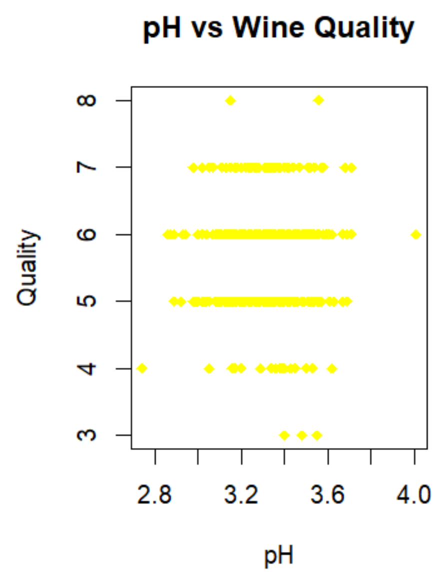
Relation of Each Variable with Quality



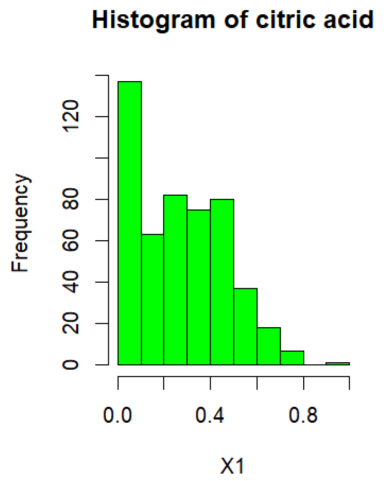




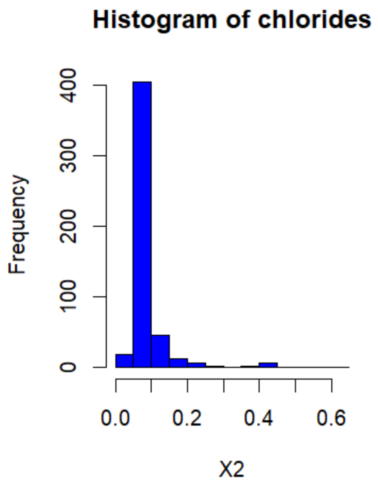




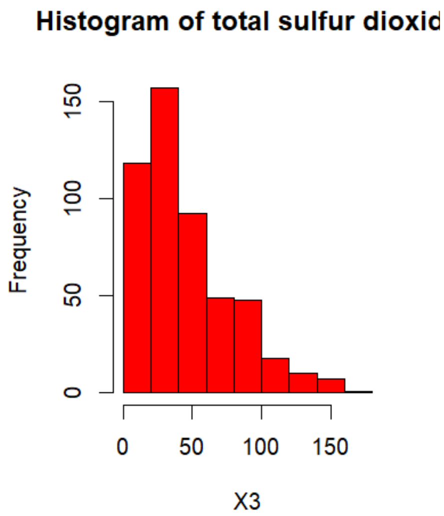
Data Distribution of Each Variable



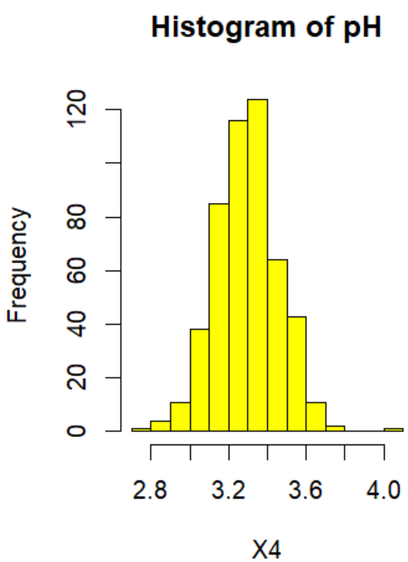
The histogram of citric acid (X1) shows a right-skewed distribution. Most wine samples have a lower citric acid content, with a high frequency of wines in the range closer to 0. This indicates that higher citric acid levels are less common in this dataset. The tail of the distribution extending towards higher citric acid values reflects that while there are wines with such characteristics, they are relatively rare.



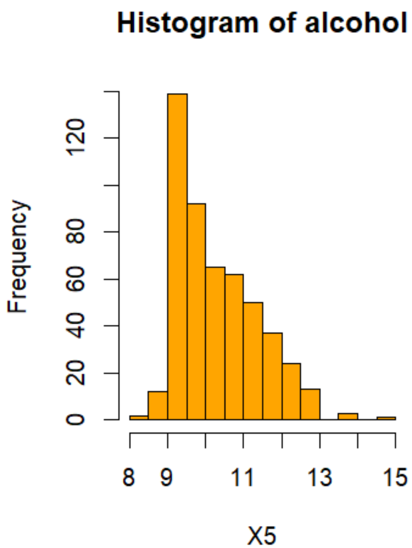
The histogram for chlorides (X2) in the wine samples shows a pronounced right-skewed distribution, with a steep drop-off in frequency as chloride levels increase. The bulk of wines have chlorides concentrated at the lower end of the scale, with very few instances of high chloride levels. This pattern indicates a heavy preference or natural occurrence of lower chloride concentrations in the sampled wines



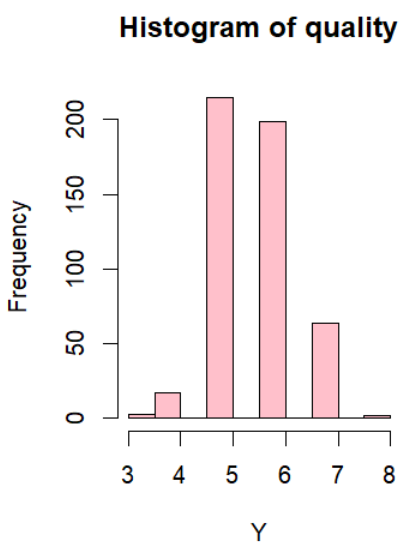
The histogram for total sulfur dioxide (X3) presents a right-skewed distribution. Most wine samples have a lower concentration of total sulfur dioxide, with frequencies diminishing as levels increase. The data indicates a commonality of wines with sulfur dioxide levels clustered in the lower range, while higher levels are less typical among these samples.



The pH histogram (X4) shows a roughly symmetrical distribution centered around a mid-range pH value, with the frequency tapering off more gradually on both the lower and higher ends of the pH scale. This distribution suggests that the pH levels in most wine samples are concentrated around a central value, with fewer instances of very high or very low pH values.

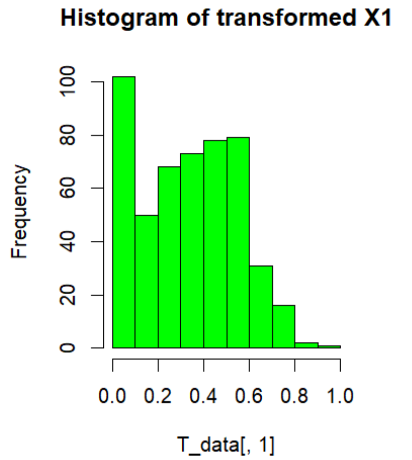


The histogram for alcohol content (X5) shows a right-skewed distribution. The majority of wine samples have a moderate alcohol percentage, with the frequency progressively decreasing as the alcohol content increases. This distribution indicates that wines with lower to moderate alcohol levels are more common in this sample set, while wines with very high alcohol content are less frequent. The highest frequency occurs in the range just below 10% alcohol, suggesting this is the most typical alcohol content among these wines

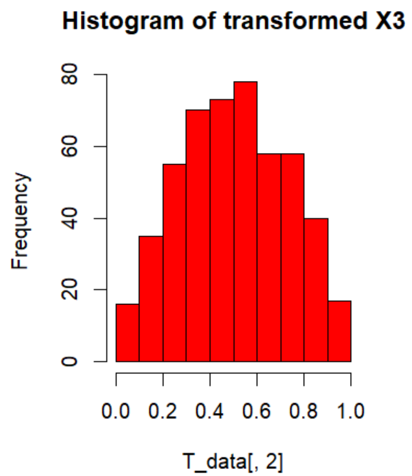


The histogram for wine quality (Y) indicates that the distribution of quality ratings is concentrated around the middle scores, with the highest frequency at the median quality score levels. There are very few wines at the extreme ends of the quality scale, suggesting that most wines fall within the average quality range, with exceptional qualities (both high and low) being less common

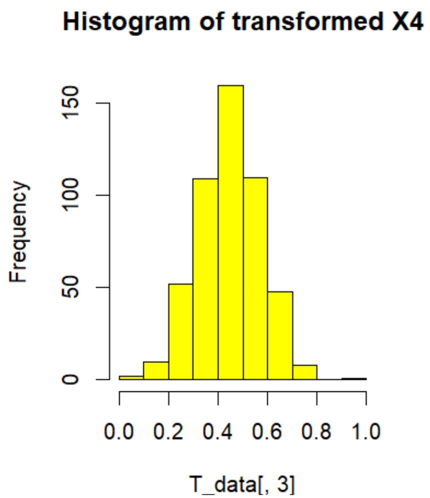
Data transformation applied(Excluding Chlorides)



X1 – Polynomial transformation and scaling
The variable X1 had a skewness of **0.358**. The data was transformed using a polynomial transformation where the p value was equal to 0.8. After the transformation the skewness value reduced to **0.066**. This means the data is now more near to normal distribution. Then in order to further normalise, the data was scaled using a min-max function. Now, the data values are all in between the range of **0 to1**

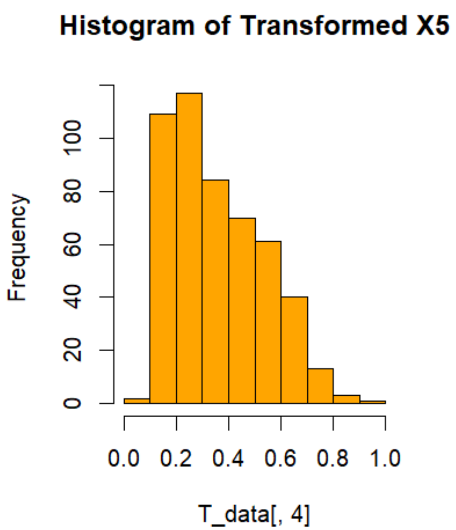


X3- Log transformation and scaling
The variable X3 had a skewness of **1.172**. The data was transformed using a log transformation After the transformation the skewness value reduced to **0.019**. This means the data is now more near to normal distribution. Then in order to further normalise, the data was scaled using a min-max function. Now, the data values are all in between the range of **0 to1**



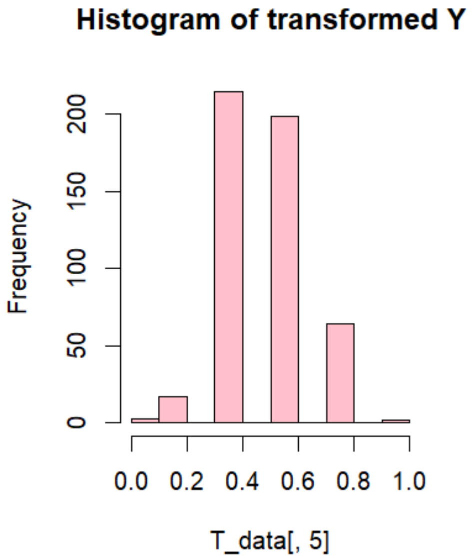
X4- Scaling

The variable X4 had a skewness of **0.041**. The data is already near to a normal distribution. Then, in order to further normalise, the data was scaled using a min-max function. Now, the data values are all in between the range of **0 to1**



X5- Log transformation and scaling

The variable X2 had a skewness of **0.842**. The data was transformed using a log transformation. After the transformation the skewness value reduced to **0.638**. This means the data is now more near to normal distribution. Then in order to further normalise, the data was scaled using a min-max function. Now, the data values are all in between the range of **0 to1**



Y- Scaling

The variable X4 had a skewness of **0.0116**. The data is already near to a normal distribution. Then, in order to further normalise, the data was scaled using a min-max function. Now, the data values are all in between the range of **0 to1**

Error measures and correlation coefficient

	Weighted Arithmetic Mean	Weighted Power Mean(p=0.5)	Weighted Power Mean(p=2)	Ordered Weighted Mean
RMSE	0.190356502257122	0.198455975322432	0.178811561432066	0.170954907888638
Av. Abs error	0.147587038947171	0.153443133887966	0.138463810010145	0.135985904053408
Pearson Correlation	0.236712151417878	0.199606126723542	0.328434408633646	0.21714416038201
Spearman Correlation	0.236183069524956	0.189042764127948	0.338390810981981	0.188875589249627

Weights/Parameters

Weighted Arithmetic Mean	Weighted Power Mean(p=0.5)	Weighted Power Mean(p=2)	Ordered Weighted Mean
0.068951125392768	0.0132748420200906	0.142930167847207	0
0.192537599316559	0.220931695174624	0.109530540857148	0.195732854699964
0.429186170721502	0.485434880639516	0.325987049795178	0.589288575964946
0.309325104569171	0.280358582165769	0.421552241500468	0.214978569335089

Model description

The Weighted Power Mean (WPM) with p=2 outperforms other models with the lowest RMSE (0.179) and highest correlation coefficients (0.328 Pearson, 0.338 Spearman), indicating strong predictive accuracy. The Ordered Weighted Averaging (OWA) model also shows good accuracy with a low RMSE (0.171). In contrast, the WPM with p=0.5 and the Weighted Arithmetic Mean (WAM) have higher RMSEs and weaker correlations, demonstrating lesser predictive capabilities.

Importance of each variable

X1 - Citric Acid:

Importance: Citric acid is one of the key organic acids found in wines. It contributes to the wine's acidity, which is essential for the balance of a wine. Acidity affects the taste, color, and microbial stability of the wine. In red wines, citric acid can add freshness and is sometimes added to wines that lack vivacity and are too flat.

Impact on Quality: The right balance of citric acid can enhance the perceived quality of red wine by contributing to the complexity of its flavour profile and stability.

X3 - Total Sulphur Dioxide:

Importance: Sulphur dioxide (SO2) acts as an antioxidant and a preservative in wine. It helps in preventing oxidative spoilage and maintaining freshness. It also inhibits the growth of microbes that could spoil the wine.

Impact on Quality: Too little sulphur dioxide, and the wine may oxidize and spoil; too much, and it can negatively impact the wine's taste and aroma. The appropriate amount of SO2 can therefore be crucial to the perceived quality of red wine.

X4 - pH:

Importance: The pH level in wine is a key factor that influences the colour, stability, taste, and aging potential. It can affect the effectiveness of sulphur dioxide as a preservative.

Impact on Quality: Wines with the optimal pH are more stable and tend to have a brighter color and fresher flavours. Wines with high pH may taste flat and are more prone to bacterial spoilage, while wines with too low pH might be overly tart.

X5 – Alcohol:

Importance: Alcohol is a by-product of fermentation and contributes to the body, flavor, and mouth feel of the wine.

Impact on Quality: Alcohol content needs to be in balance with other wine components for the wine to be considered high quality. Too much alcohol can overwhelm the flavor and make the wine feel heavy or 'hot', while too little can make it feel weak or watery

Best fitting Function

Based on the data, the **WPM with p=2** model is the best for prediction. It has the lowest RMSE, suggesting the smallest average errors, and the highest correlation coefficients, indicating the strongest relationships with actual observed data. The **OWA** model follows closely in terms of RMSE and has the lowest average absolute error, but its correlation coefficients are weaker compared to the WPM with p=2 model, making it slightly less desirable in terms of both error metrics and the strength of the relationship to the actual data

Prediction Result

The new variables are transformed as corresponding to the data trained for building the model and scaled. The variable and weights are assigned to the WPM model and a Y value is predicted. In order to get the correct quality the Y value is re-scaled to the original value

Predicted Wine Quality : 5.201294 rounded to **5**

The model has shown excellent predictive performance based on key metrics. With the lowest RMSE and highest correlation coefficients among the models tested, it effectively predicts outcomes that closely match the actual observed values. Within the context of wine quality scores ranging from 3 to 8, the model's prediction of about 5 situates the wine squarely in the middle of this spectrum, indicating average quality. This outcome is consistent with the dataset used to design the model. Hence, a predicted quality score of 5 is both reasonable and indicative of moderate quality, aligning well with the expectations set by the dataset's range.

Optimal conditions

To potentially achieve the highest quality of red wine which is 8, the optimal conditions would be ,

- Citric acid would be close to 0.4
- Chlorides would be around 0.07 to 0.08
- Total Sulphur Dioxide would be in a moderate range around 37 to 46
- pH would ideally be around 3.3 to 3.4
- V5 would be exceeding 10.0